# METAL FORMING PROCESSES

*Hot and Cold Working • Rolling • Extrusion • Drawing • Tube Making • Forging*

**Reference: Elements of Workshop Technology, Vol. II - Hajra Choudhury**

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## 1. Introduction to Metal Forming

Metal forming is the shaping of metals by plastic deformation under the action of externally applied forces. Unlike machining, the desired shape is obtained mainly by movement of material rather than by removal of chips. The volume of metal is approximately conserved, while its dimensions and internal structure change.

- Bulk deformation processes: rolling, forging, extrusion, wire/rod drawing and tube drawing.
- Sheet-metal processes: bending, deep drawing, shearing, spinning and related operations. The present syllabus focuses mainly on bulk metal forming.

The ability of a material to undergo plastic deformation without cracking depends on composition, temperature, strain rate, state of stress, grain size, lubrication and previous deformation history.

## 2. Hot Working and Cold Working

### 2.1 Principle of hot working

Hot working is plastic deformation carried out above the recrystallization temperature of the metal and below the melting temperature. During deformation, new strain-free grains are continuously formed or restored by recovery and recrystallization. Therefore the metal does not accumulate severe strain hardening.

Typical hot-working operations include hot rolling of blooms and billets, hot forging, hot extrusion and hot piercing of tubes.

- Lower flow stress and lower forming load than cold working.
- Large deformation can be given without cracking in many metals.
- Cast structure is broken down; internal voids may be reduced and grains are refined.
- No appreciable work hardening remains after recrystallization.

## 2.2 Principle of cold working

Cold working is plastic deformation performed below the recrystallization temperature, usually at or near room temperature. The metal becomes progressively strain hardened: strength and hardness increase while ductility decreases. Recrystallization does not occur during the operation, so deformation effects remain in the microstructure.

Common cold-working processes are cold rolling, cold extrusion, wire drawing, tube drawing, cold heading and cold forging.

## 2.3 Classification

| Basis | Hot working | Cold working |
|---|---|---|
| Temperature | Above recrystallization temperature | Below recrystallization temperature |
| Typical processes | Hot rolling, forging, extrusion, piercing | Cold rolling, wire/tube drawing, cold extrusion, heading |
| Dominant metallurgical feature | Recovery/recrystallization during deformation | Strain hardening |
| Usual objective | Large shape change and structure improvement | Accurate size, finish and strengthening |

## 2.4 Properties and effects

| Feature | Hot working | Cold working |
|---|---|---|
| Strength and hardness | Generally lower after working because recrystallization removes work hardening | Increase due to strain hardening |
| Ductility | Generally retained or improved | Decreases with increasing cold reduction |
| Residual stress | Low compared with cold working | May be significant |
| Grain structure | Refined, recrystallized grains; cast dendritic structure broken | Elongated/deformed grains and preferred orientation |
| Surface quality | Scale and oxidation can impair finish | Good surface finish possible |
| Dimensional accuracy | Moderate because of thermal contraction/scale | High |
| Force required | Lower | Higher |

## 2.5 Advantages and disadvantages

Advantages of hot working:

- Large deformation is possible with comparatively low force.
- Porosity and internal cavities can be reduced by compressive working.
- Grain refinement improves uniformity and mechanical properties.
- Segregated or coarse cast structure is broken down.
- Residual stresses are small and strain hardening is avoided.

Disadvantages / limitations of hot working:

- Oxidation and scale formation cause metal loss and poor finish.

- Close dimensional tolerances are difficult because of thermal contraction.
- Heating equipment and temperature control are required.
- Tool life may be reduced by high temperature and thermal cycling.
- Decarburization may occur in steels under unsuitable furnace conditions.

Advantages of cold working:

- Excellent surface finish and dimensional accuracy.
- Strength and hardness increase by strain hardening.
- No heating cost for the workpiece in ordinary cold working.
- Close tolerances and thin sections are possible.
- Good reproducibility in mass production.

Disadvantages / limitations of cold working:

- Higher forces and more powerful equipment are required.
- Ductility decreases and cracking may occur after excessive reduction.
- Residual stresses and anisotropy may develop.
- Intermediate annealing may be needed between large reductions.
- Very large shape changes are more difficult than in hot working.

### 2.6 Difference between hot working and cold working

| Point | Hot working | Cold working |
|---|---|---|
| Working temperature | Above recrystallization temperature | Below recrystallization temperature |
| Deformation resistance | Low | High |
| Work hardening | Removed by recrystallization | Pronounced |
| Ductility during process | High | Progressively decreases |
| Surface finish | Relatively poor because of scale | Good |
| Dimensional accuracy | Lower | Higher |
| Residual stress | Small | Higher |
| Grain form | Equiaxed recrystallized grains | Elongated/distorted grains |
| Typical use | Primary breakdown and large sections | Finishing, strengthening, precision sizing |

## 3. Rolling

### 3.1 Principle of rolling

Rolling is a metal-forming process in which the workpiece is passed between rotating rolls. Friction between roll and metal draws the work into the roll gap, while compressive forces reduce thickness or alter cross-section. Rolling may be carried out hot or cold.

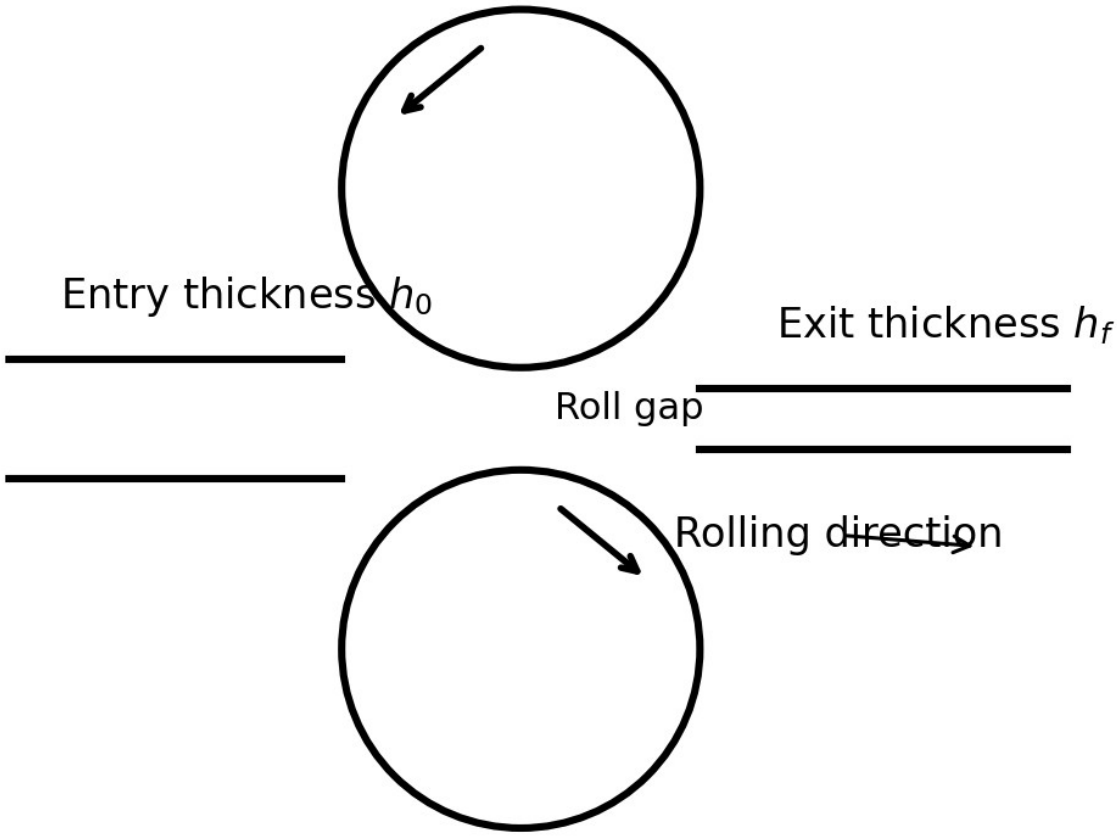



*Figure 1. Basic principle and geometry of flat rolling.*

### 3.2 Important rolling parameters

Let h0 be the entry thickness, $h_f$ the exit thickness, b the strip width, R the roll radius, and v the strip velocity.

$$\textbf{Draft, } d = h_o - h_f$$

$$\textbf{Reduction (\%)} = [(h_o - h_f) / h_o] \times 100$$

$$\textbf{True strain in thickness direction, } \varepsilon = \ln(h_o / h_f)$$

The contact length for small reductions in flat rolling is approximately:

$$L \approx \sqrt{(R\,\Delta h)}, \text{ where } \Delta h = h_o - h_f$$

The maximum possible draft is governed by friction. A commonly used limiting relation is:

$$\Delta h_{max} \approx \mu^2 R$$

where μ is the coefficient of friction between roll and work. Thus larger rolls and higher friction permit a greater bite, though excessive friction increases power and roll wear.

### 3.3 Rolling process variables

- Roll diameter: larger rolls increase contact length and rolling load; small work rolls are therefore supported by larger backup rolls in high-precision mills.

- Coefficient of friction and lubrication: enough friction is required to bite the strip, but excessive friction raises load and power.
- Draft/reduction per pass: controls deformation, load, number of passes and final properties.
- Rolling speed: affects productivity, temperature and friction conditions.
- Entry and exit tension: reduce rolling load and improve strip control; excessive tension can cause breakage.
- Material flow stress: higher strength or lower temperature increases rolling force.
- Roll crown and elastic deflection: influence final thickness distribution and flatness.
- Temperature: hot rolling lowers flow stress; cold rolling gives better accuracy and finish.

## 3.4 Types of rolling mills

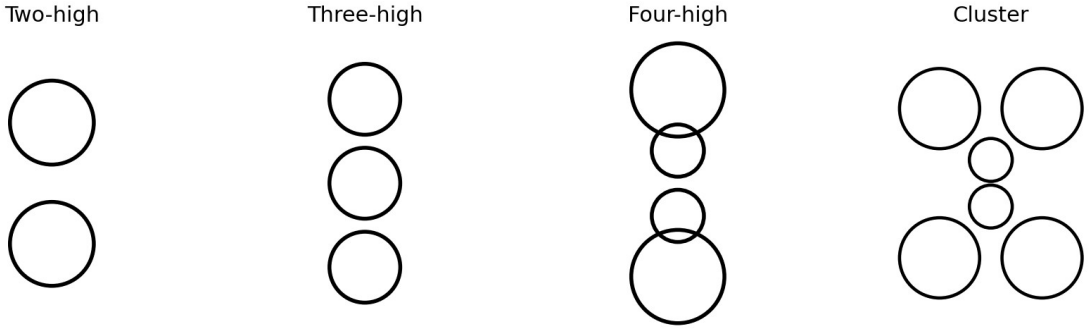



*Figure 2. Common rolling-mill arrangements.*

| Mill type | Construction / feature | Typical use |
|---|---|---|
| Two-high | Two opposed rolls; may be non-reversing or reversing | Blooming, roughing, plate and simple rolling |
| Three-high | Three rolls; work passes alternately in opposite directions without reversing roll rotation | Repeated hot rolling passes |
| Four-high | Small work rolls supported by large backup rolls | Thin sheet and strip, especially cold rolling |
| Cluster / Sendzimir | Very small work rolls supported by several backup rolls | Very thin and hard strip |
| Tandem / continuous | Several rolling stands in series | High-production strip and sheet |
| Universal mill | Horizontal plus vertical rolls | Rails, structural sections, wide-flange beams |

## 3.5 Rolling passes

A rolling pass is one passage of metal through a particular roll gap or groove. When a large reduction or major change of section is required, the work is passed through a sequence of roll grooves called a pass schedule or pass design.

- Breakdown passes reduce a large ingot or billet to a manageable section.
- Roughing passes provide major reduction and approximate shape.
- Intermediate passes progressively reshape the section.
- Finishing passes give final dimensions, contour and surface quality.

For sections such as rounds, squares, rails and channels, successive grooves are designed to maintain stable metal flow and avoid folds, overfilling and excessive spread.

### 3.6 Rolling defects and remedies

| Defect | Cause | Remedy |
|---|---|---|
| Wavy edges | Roll bending causes greater reduction at centre than edges | Use roll crown/backup rolls; control roll bending and tension |
| Centre buckle | Edges are reduced more than centre | Correct crown, alignment and tension distribution |
| Edge cracks | Low ductility, excessive reduction, improper temperature | Reduce draft; control temperature; improve material ductility |
| Zipper / centre cracks | High tensile stress at strip centre, poor reduction schedule | Optimize draft, crown and material condition |
| Alligatoring | Non-uniform deformation or internal defects cause slab to split along thickness | Improve ingot quality, temperature uniformity and pass schedule |
| Scale / rolled-in oxide | Oxide scale not removed before rolling | Descale effectively; control furnace atmosphere |
| Thickness variation | Roll deflection, eccentricity, thermal crown, poor gap control | Use automatic gauge control, roll grinding and backup rolls |

### 3.7 Applications of rolling

- Production of plates, sheets, strips and foils.
- Bars, rods and wire-rod stock.
- Structural sections such as angles, channels, I-sections and rails.
- Rings and railway wheels by specialized rolling processes.
- Threads and gears by roll-forming methods.
- Primary conversion of ingots/slabs into blooms, billets and slabs.

## 4. Extrusion

### 4.1 Description and principle

Extrusion is a compressive forming process in which a billet is forced to flow through a die opening so that the emerging product has approximately the same cross-section as the die aperture. It is especially suitable for long products of constant but complex cross-section.

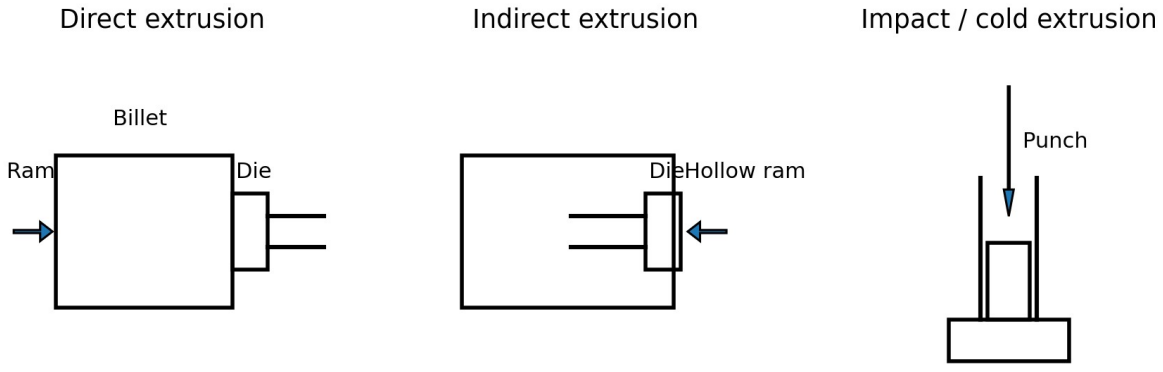



*Figure 3. Direct, indirect and impact/cold extrusion arrangements.*

### 4.2 Hot and cold extrusion

| Feature | Hot extrusion | Cold extrusion |
|---|---|---|
| Temperature | Above recrystallization temperature | At or near room temperature |

| | | |
|---|---|---|
| Force | Lower because flow stress is reduced | Higher |
| Surface finish | May be affected by oxidation/scale | Very good |
| Accuracy | Moderate | High |
| Reduction possible | Large | Limited by force and ductility |
| Typical products | Large sections, difficult alloys, Al/Mg/Cu sections | Fasteners, collapsible tubes, small precision parts |

## 4.3 Types of extrusion

**Direct or forward extrusion:** The ram pushes the billet toward a stationary die. Metal flow is in the same direction as ram motion. Billet-container friction is significant.

**Indirect or backward extrusion:** The die moves into the billet while the product flows opposite to the die/ram motion. Relative sliding between billet and container is reduced, so extrusion force is lower.

**Hydrostatic extrusion:** The billet is surrounded by pressurized fluid and driven through the die. Very high pressure and low friction make it useful for brittle or difficult materials.

**Impact extrusion:** A punch strikes a slug at high speed, causing metal to flow forward, backward or both. Commonly a cold-working operation for thin-walled components.

**Lateral / side extrusion:** Material flows through an opening located sideways relative to ram motion.

## 4.4 Advantages, limitations and applications

Advantages:

- Complex solid and hollow sections can be produced in one operation.
- Good material utilization with little scrap.
- Long continuous lengths are possible.
- Hot extrusion can form alloys that are difficult to roll or forge.
- Cold extrusion gives good finish, high accuracy and beneficial work hardening.
- Grain flow is generally favorable along the product length.

Limitations / disadvantages:

- High die and press cost.
- Extrusion force may be very large, particularly in cold extrusion.
- Die wear and thermal loading can be severe.
- Cross-section is essentially constant along length in ordinary extrusion.
- Surface cracking, pipe and internal cracking can occur if process variables are poorly controlled.
- Lubrication is critical, especially in hot extrusion of steels.

Applications:

- Aluminium architectural sections, window frames and channels.
- Tubes and hollow sections.
- Heat-sink profiles and electrical bus-bar shapes.
- Automotive and aircraft structural sections.
- Collapsible tubes, cans and small cold-extruded components.
- Rods and bars of non-ferrous alloys and steels.

# 5. Wire, Rod and Tube Drawing

## 5.1 Principle of drawing

Drawing is a tensile-compressive metal-forming process in which a rod, wire or tube is pulled through a converging die. The cross-sectional area decreases and the length increases. Unlike extrusion, the driving force is applied to the product on the exit side.

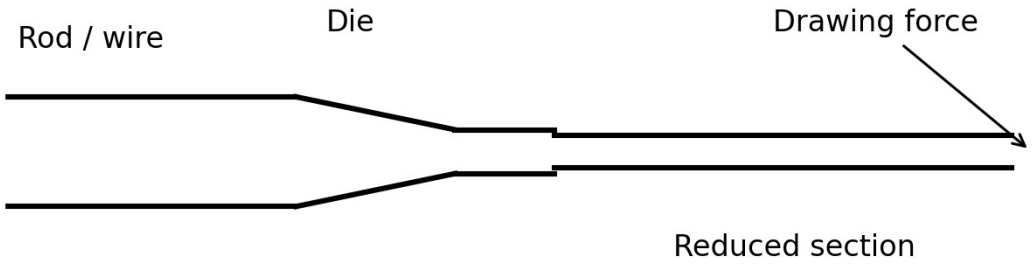



*Figure 4. Principle of wire/rod drawing through a die.*

## 5.2 Wire drawing

Wire drawing starts with wire rod, which is cleaned to remove scale, coated or lubricated, pointed at one end, threaded through a die and gripped by a drawing block. Because a large reduction cannot usually be made in one pass, several dies of progressively smaller diameter are used.

1. Clean and descale the rod by pickling, mechanical descaling or shot blasting.
2. Apply lubricant or a carrier coating such as lime/phosphate where appropriate.
3. Point the leading end so that it can pass through the first die.
4. Pull through one or more dies; each pass gives a controlled reduction.
5. Use intermediate annealing if excessive strain hardening would otherwise cause fracture.

Important variables include die angle, reduction per pass, friction/lubrication, drawing speed, material strength and back tension.

## 5.3 Rod drawing

Rod drawing is similar to wire drawing but is generally used for larger diameters and shorter lengths. Draw benches are commonly used. The process improves dimensional accuracy and surface finish and is often used to produce bright steel bars.

## 5.4 Tube drawing

Tube drawing reduces diameter and/or wall thickness. Depending on the required control of internal diameter and wall thickness, the tube may be drawn without an internal tool, over a fixed mandrel, over a floating plug, or over a long mandrel.

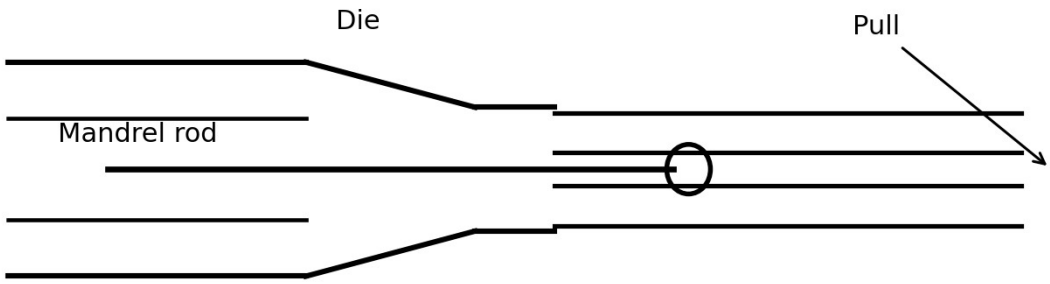



*Figure 5. Tube drawing through a die with internal mandrel control.*

| Method | Internal tool | Main effect |
|---|---|---|
| Tube sinking | None | Reduces outside diameter; wall thickness control is limited |
| Fixed-plug / fixed-mandrel drawing | Stationary plug or mandrel | Good control of inside diameter and wall thickness |
| Floating-plug drawing | Plug floats in working zone | Suitable for long lengths; good wall control |
| Draw over mandrel (DOM) | Mandrel extends through die zone | Excellent dimensional control and surface finish |

## 6. Tube Making

Tubes are manufactured either by seamless methods or by forming strip and joining its edges. The choice depends on diameter, wall thickness, material, pressure rating and production quantity.

### 6.1 Seamless tube making

A common route for seamless steel tubes is rotary piercing (Mannesmann process). A heated round billet is rotated and compressed between skewed rolls. Because of the stress state and internal tensile effect, a central cavity forms; a piercing plug enlarges and controls the hole. The hollow shell is then elongated and sized by plug mills, mandrel mills or stretch-reducing mills.

- No longitudinal weld seam, making the process attractive for high-pressure and critical service.
- Subsequent drawing can improve dimensional accuracy and surface finish.

### 6.2 Welded tube making

In welded-tube manufacture, strip or plate is progressively formed into a cylindrical shape and the edges are joined. Depending on size and product, joining may be by resistance welding, high-frequency welding, submerged-arc welding or other methods. Weld bead treatment, sizing and inspection follow.

- Electric-resistance/high-frequency welded tubes are widely used for smaller and medium diameters.
- Submerged-arc welded pipe may be longitudinally or spirally formed for large diameters.

## 7. Difference between Extrusion and Drawing

| Point | Extrusion | Drawing |
|---|---|---|
| Primary driving stress | Compressive | Tensile pull with compressive die pressure |
| Material movement | Billet is pushed through die | Product is pulled through die |
| Starting stock | Billet or slug | Rod, wire or tube |
| Typical products | Solid/hollow sections, complex profiles | Wire, rods, bars and tubes |
| Cross-sectional complexity | High; complex sections possible | Usually simple sections, especially round |
| Reduction in one operation | Often large, especially hot extrusion | Usually limited; multiple passes common |
| Friction characteristic | Container-wall friction important in direct extrusion | Die friction important; no billet container |
| Product length | Limited by billet/press arrangement in batch extrusion | Long continuous lengths possible, particularly wire |
| Hot/cold use | Both hot and cold widely used | Mostly cold, though warm/hot variants exist |

## 8. Forging

### 8.1 Principle and characteristics

Forging is the shaping of metal by localized compressive forces applied through hammers, presses or dies. It may be performed hot, warm or cold. Hot forging is most common for large shape changes because the metal has high ductility and low flow stress at forging temperature.

A major advantage of forging is the controlled flow of grains or fibre lines around the shape, which can provide high strength, toughness and fatigue resistance when the process and die design are appropriate.

### 8.2 Forgeability

Forgeability is the ability of a material to undergo deformation in forging without cracking or developing unacceptable defects. It depends on material composition, microstructure, temperature, strain rate, stress state and amount of deformation.

- High ductility and low flow stress generally indicate good forgeability.
- Compressive hydrostatic stress improves forgeability; tensile stress promotes cracking.
- Improper temperature can reduce ductility and cause hot shortness or cold cracking.

### 8.3 Forging temperature

The forging temperature range must be high enough to provide adequate ductility and low deformation resistance, but below temperatures that cause burning, excessive oxidation or severe grain growth. The work should normally be completed above the lower critical forging temperature to avoid cracks and excessive tool loads.

| Material (typical) | Approximate hot-forging range* |
|---|---|
| Low-carbon steel | About 1250°C down to 850–900°C |
| Medium/high-carbon steel | Generally somewhat lower upper temperature than mild steel |

| | |
|---|---|
| Aluminium alloys | Roughly 350–500°C depending on alloy |
| Copper alloys | Commonly several hundred °C; range depends strongly on alloy |
| Titanium alloys | High-temperature controlled-atmosphere forging; exact range alloy-specific |

*Ranges are indicative only; actual practice depends on composition and product specification.

## 8.4 Forgeable materials

Many steels, aluminium alloys, copper alloys, magnesium alloys, nickel alloys and titanium alloys are forged. Low- and medium-carbon steels generally possess good hot forgeability. Cast irons are not ordinarily forgeable because of their low ductility, and high-alloy or high-carbon materials require stricter temperature control.

## 8.5 Furnaces used for forging

| Furnace / heating method | Features / use |
|---|---|
| Coal or coke-fired forge | Traditional small-scale heating; simple but less uniform and more polluting |
| Oil-fired furnace | Good heating rate; used in many forge shops |
| Gas-fired furnace | Clean, controllable and widely used for batch/continuous heating |
| Electric resistance furnace | Accurate temperature control and clean atmosphere |
| Induction heating | Very rapid, localized and controllable heating; excellent for production forging |
| Salt-bath / special atmosphere | Used where oxidation control or special heating conditions are needed |

## 8.6 Forging tools

- Anvil: supporting surface for hand forging and smith work.
- Hammers: hand hammer, sledge hammer, power hammer and drop hammer for delivering blows.
- Tongs: hold and manipulate hot workpieces.
- Fullers: spread metal locally and aid drawing out.
- Flatters: smooth and finish flat surfaces.
- Swages: form or finish round, hexagonal or shaped sections.
- Set hammer: produces shoulders and local changes of section.
- Punches and drifts: create and size holes.
- Chisels / hot sets: cut hot metal.
- Heading tools and dies: form heads, bosses and upset regions.

## 8.7 Common forging operations

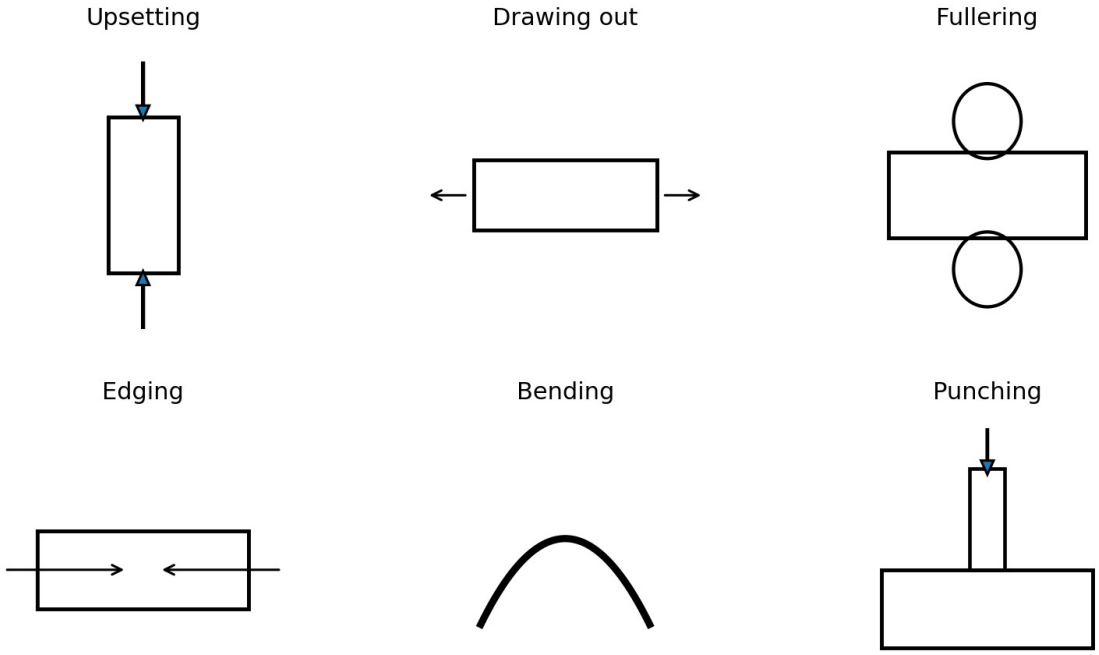



*Figure 6. Typical hand/press-forging operations.*

| Operation | Purpose |
|---|---|
| Upsetting | Increases cross-sectional area by reducing length |
| Drawing out / drawing down | Reduces cross-section and increases length |
| Fullering | Spreads metal away from a localized region and prepares stock for drawing |
| Edging | Gathers metal into a required region, often before impression-die forging |
| Bending | Changes direction or curvature of the workpiece |
| Punching | Produces a hole by driving a punch through hot metal |
| Drifting | Enlarges/shapes a previously punched hole |
| Cutting | Separates excess stock or cuts to length |
| Swaging | Reduces or finishes round/polygonal sections using shaped tools |

## 8.8 Classification of forging

- According to temperature: hot forging, warm forging, cold forging.
- According to equipment: hand/smith forging, hammer forging, press forging, upset forging, roll forging, swaging.
- According to die constraint: open-die forging, impression-die forging and closed-die/flashless forging.

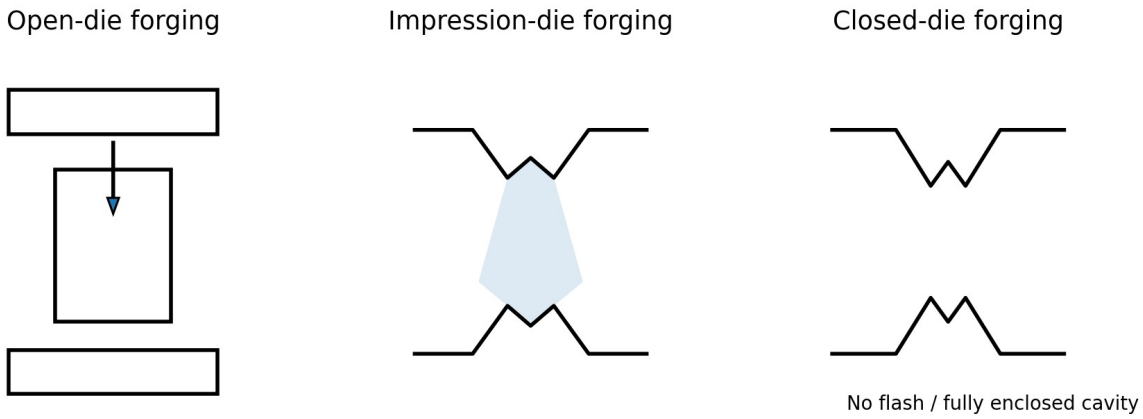



*Figure 7. Open-die, impression-die and closed-die forging concepts.*

## 8.9 Open-die forging

In open-die forging, the work is compressed between simple dies that do not completely enclose the metal. Material is free to flow laterally. The work is repeatedly repositioned to obtain the required shape.

- Used for large shafts, discs, rings and low-quantity components.
- Low die cost and great flexibility.
- Lower dimensional accuracy; skilled manipulation and subsequent machining are often required.

## 8.10 Impression-die forging

In impression-die forging, shaped die cavities partially enclose the work. Excess metal normally flows into a narrow flash gutter around the die parting line. The flash cools rapidly and offers high resistance to flow, helping force metal to fill the cavity.

- Suitable for medium-to-high production of connecting rods, crankshafts, hand tools and similar parts.
- Usually involves blocker and finisher impressions, followed by flash trimming.

## 8.11 Closed-die / flashless forging

In closed-die or flashless forging, the billet is fully enclosed and essentially no flash is permitted. Billet volume and positioning must be controlled accurately; otherwise incomplete filling or excessive die pressure can occur.

- Very high material utilization and good dimensional control.
- Requires accurate stock volume, close process control and costly tooling.

## 8.12 Forging defects and remedies

| Defect | Likely cause | Remedy |
|---|---|---|
| Unfilled section / underfill | Insufficient stock, low temperature, poor die design or inadequate force | Increase stock/temperature/energy; improve preform and die filling |
| Cold shut | Two metal streams meet but do not weld because of folds or oxide | Improve preform, radii and temperature; avoid sharp flow reversal |
| Lap / fold | Improper material flow, sharp corners or oversized stock | Improve die design and stock distribution; provide proper fillets |
| Scale pits | Oxide scale pressed into surface | Descale billet/dies; control furnace atmosphere and heating time |
| Die shift / mismatch | Misalignment of upper and lower dies | Align guides and dies; maintain equipment |

| | | |
|---|---|---|
| Internal cracks / flakes | Excessive tensile stress, low ductility, improper reduction or cooling | Control deformation, temperature and cooling; improve stock quality |
| Excessive flash | Too much stock or improper die closure | Control billet volume and die setting |
| Distortion / warpage | Non-uniform cooling or residual stress | Use controlled cooling, straightening and proper heat treatment |

### 8.13 Advantages of forging

- High strength and toughness due to refined grain structure and favorable grain flow.
- Internal voids and porosity can be reduced in hot forging.
- Excellent reliability for dynamically loaded and safety-critical parts.
- Broad range of sizes from small fasteners to very large shafts and discs.
- Near-net-shape production is possible in impression and precision forging.
- Good material utilization compared with machining from solid stock.

### 8.14 Disadvantages and limitations of forging

- High die and equipment cost for impression/closed-die forging.
- Oxidation, scale and decarburization in hot forging.
- Dimensional tolerances are generally poorer than machining unless precision forging is used.
- Complex thin ribs, undercuts and very intricate shapes are difficult.
- Noise and vibration can be high with hammers.
- Large forging loads require heavy foundations and robust machinery.
- Finishing, trimming, heat treatment and machining may still be required.

### 8.15 Applications of forging

- Crankshafts, connecting rods, gears and transmission components.
- Aircraft landing-gear parts and highly stressed structural fittings.
- Bolts, rivets, fasteners and hand tools.
- Turbine discs, shafts and rings.
- Railway and automotive axles.
- Valves, pipe fittings, hooks, chains and lifting components.

## 9. Important Points

- Hot working is carried out above recrystallization temperature; cold working is below it.
- Cold working improves strength, hardness, accuracy and finish but reduces ductility and raises forming force.
- Rolling reduces thickness by compressive forces between rotating rolls; friction provides the bite.
- Draft = $h_0 - h_f$; maximum possible draft increases with coefficient of friction and roll radius.
- Four-high and cluster mills use small work rolls supported by larger backup rolls for thin strip.
- Extrusion pushes a billet through a die; drawing pulls rod, wire or tube through a die.
- Direct extrusion has billet-container sliding friction; indirect extrusion greatly reduces this friction.
- Wire drawing normally uses several dies and may require intermediate annealing.

- Tube drawing methods include sinking, fixed plug/mandrel, floating plug and draw-over-mandrel.
- Seamless tubes can be produced by rotary piercing followed by elongation and sizing.
- Forging uses compressive forces and can improve grain flow, toughness and fatigue resistance.
- Open-die forging uses simple dies; impression-die forging usually produces flash; closed-die forging aims at flashless filling.
- Common forging defects include laps, cold shuts, underfill, scale pits, mismatch and internal cracks.

## 10. Important Questions

1. Explain the principles of hot working and cold working. Compare them in tabular form.
2. Discuss the advantages, disadvantages and metallurgical effects of hot and cold working.
3. Explain the principle of rolling and derive/define draft, reduction and contact length.
4. Classify rolling mills with neat sketches and state their applications.
5. Explain rolling passes, rolling variables and common rolling defects with remedies.
6. Describe direct and indirect extrusion. Compare hot and cold extrusion.
7. Explain wire drawing and tube drawing with sketches.
8. Describe methods of seamless and welded tube manufacture.
9. Differentiate between extrusion and drawing.
10. Define forgeability. Explain factors affecting forgeability and forging temperature.
11. List and explain common forging tools and forging operations.
12. Compare open-die, impression-die and closed-die forging.
13. Explain forging defects and their remedies.
14. Write the advantages, disadvantages, limitations and applications of forging.

## Reference

Hajra Choudhury, S. K. and Hajra Choudhury, A. K., Elements of Workshop Technology, Vol. II, Media Promoters & Publishers Pvt. Ltd. The present notes are organized around the standard workshop-technology treatment of metal-forming processes and are written in original explanatory language.